

Kinematics Enable Device-Agnostic Biological Joint Moment Estimation

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Abstract—Objective: Data-driven models that estimate physiological states, particularly biological joint moments, are widely used in exoskeleton control. However, these estimators are often coupled to device-specific sensor configurations, limiting controller transfer and the use of open-source biomechanics datasets. **Methods:** We proposed joint kinematics as an intermediate representation that decouples hardware-specific sensing from downstream biological joint moment estimation. A joint moment estimator using joint angles and angular velocities was trained exclusively on open-source biomechanics data and evaluated using kinematics from a hip exoskeleton, knee exoskeleton, and Xsens Awinda IMU system during level-ground, ramp-ascent, and ramp-descent walking. **Results:** The estimator achieved an RMSE of 0.169 ± 0.067 Nm/kg and R^2 of 0.801 ± 0.088 using hip-exoskeleton kinematics, 0.192 ± 0.039 Nm/kg and 0.615 ± 0.141 using knee-exoskeleton kinematics, and 0.144 ± 0.061 Nm/kg and 0.858 ± 0.121 using IMU-derived kinematics across bilateral hip, knee, and ankle joints. Condition-averaged R^2 remained at or above 0.755 for the hip exoskeleton and 0.802 for the IMU sensor suite, while knee exoskeleton performance was lower during ramp descent than ascent. **Conclusion:** Joint kinematics enabled an estimator trained only on open-source data to operate across distinct wearable platforms. **Significance:** This framework may reduce target-device data collection and support transferable biological joint moment estimation for exoskeleton control and wearable biomechanics.

Index Terms—Biological joint moment estimation, joint kinematics, device-agnostic exoskeleton control

I. INTRODUCTION

Wearable robotic systems increasingly rely on real-time estimates of the user’s biomechanical state to provide assistance that adapts across individuals and locomotor tasks [1]–[3]. Among the variables available for adaptive control, biological joint moments are particularly relevant because they quantify the net mechanical contribution generated by the user’s musculoskeletal system at each joint. Unlike discrete representations such as gait phase or terrain class [4]–[6], biological joint moments provide a continuous, mechanically interpretable measure of the user’s locomotor demand. Recent studies have demonstrated that data-driven joint moment estimates can support subject-independent and task-agnostic exoskeleton control and can provide a common basis for assistance across locomotor activities [7]–[9]. However, direct measurement of biological joint moments generally requires laboratory-based inverse dynamics using motion capture and ground-reaction

forces, motivating their estimation from wearable sensors for real-time and out-of-laboratory applications.

Existing data-driven biological joint moment estimators commonly operate directly on the sensory signals available from a particular wearable device, including inertial measurement units, motor encoders, pressure sensors, and other device-mounted measurements. These approaches can achieve accurate moment estimates when the sensing configuration used during deployment matches the configuration represented in the training data [7]–[9]. Nevertheless, the resulting input–output mapping captures not only the relationship between human movement and joint moments, but also characteristics specific to the original sensing system. Differences in sensor modality, placement, orientation, attachment compliance, calibration, joint-coordinate definitions, sampling rates, and signal processing can substantially alter the input distribution. Consequently, an estimator trained directly on one device’s signals may not be deployable on another device unless the target platform reproduces the original inputs or additional target-specific data are collected.

Recent work using deep domain adaptation has directly addressed this sensor-domain mismatch by aligning representations between source and target wearable configurations [10]. This approach represents an important advance because it can eliminate the need to collect labeled target-domain reference biomechanics, thereby avoiding repeated synchronization with motion capture and force-measurement systems. Nevertheless, domain adaptation still requires recordings from the target sensing configuration, even when those recordings are unlabeled. Its effectiveness may also depend on the signal quality, locomotor coverage, and representativeness of the collected target-domain data. Thus, changes in device hardware, sensor placement, or preprocessing may still require a new target-domain collection and adaptation procedure. These considerations motivate a complementary strategy that reduces sensor-domain mismatch by defining the interface to the downstream estimator using a common physical representation rather than directly using hardware-specific sensor signals.

The hardware dependence of raw-signal estimators also limits the broader use of open-source biomechanics data. Public datasets now provide extensive collections of lower-limb kinematics, ground-reaction forces, and inverse-dynamics joint moments across participants and locomotor conditions [11]–[14]. These resources can provide the biomechanical labels needed to train generalizable joint moment estimators and broad foundation models without requiring each research group to reproduce the corresponding laboratory data collec-

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tion [15]. However, any wearable sensor signals contained in these datasets remain tied to the modalities, placements, and processing procedures used in the original studies. Moreover, such datasets cannot contain the raw sensing representation of every custom exoskeleton or future wearable platform. Models trained directly on hardware-specific signals therefore cannot fully exploit the available kinematic and inverse-dynamics data when their required inputs are unavailable on the target device. The central bottleneck is consequently not only the availability of reference biomechanics, but also the lack of a common input representation through which those data can be reused across wearable systems.

Joint kinematics provide a potential intermediate representation for bridging device-specific sensing and generalized biological joint moment estimation. Joint angles and angular velocities can be obtained from multiple sensing configurations, including motor encoders, segment-mounted IMUs, inertial motion-capture systems, and learning-based kinematic estimators [16]–[18]. Under the proposed formulation, each wearable platform first converts its available sensor signals into joint kinematics, and a downstream estimator then maps these kinematics to biological joint moments (Fig. 1). Joint kinematics are not inherently identical across all systems, as coordinate definitions, signs, filtering, and sampling procedures must still be harmonized. Nevertheless, they provide a biomechanically interpretable physical domain that is more readily reconciled across devices than raw sensor channels. This separation allows the hardware-specific component of the pipeline to focus on producing compatible kinematic estimates, while the downstream moment estimator can be trained using open-source kinematics and inverse-dynamics labels. In contrast to adapting a model separately between each source and target sensor domain, this formulation places the reusable interface at a shared biomechanical representation. Domain adaptation may subsequently be used to refine performance when target-domain data are available, but it is not required to train the initial downstream estimator.

In this study, we investigated whether wearable-derived joint kinematics can serve as a common intermediate input domain for biological joint moment estimation across heterogeneous sensing platforms (Fig. 1). We trained causal temporal convolutional models exclusively using joint kinematics and inverse-dynamics moments from open-source biomechanics datasets and applied them, without target-device training or fine-tuning, to kinematics obtained from three distinct configurations: a bilateral robotic hip exoskeleton, a unilateral robotic knee exoskeleton, and a IMU sensor suite. We evaluated single-joint moment estimation using localized exoskeleton sensing, multi-joint estimation using whole-limb inertial kinematics, the influence of wearable-derived kinematic quality on downstream predictions, and the benefit of including adjacent-joint kinematic context. We hypothesized that harmonized joint angles and angular velocities would provide a sufficiently consistent representation to support accurate biological joint moment estimation across these disparate platforms. By decoupling hardware-specific kinematic estimation from downstream moment prediction, this framework seeks to reduce reliance on target-device data collection and facilitate the development of

modular and transferable estimators for exoskeleton control and wearable biomechanics.

II. METHODS

A. Participants

Eight healthy adults (six male, two female) participated in the study. Participants were 26.3 ± 2.3 years old, with a height of 1.74 ± 0.06 m and a body mass of 72.3 ± 13.3 kg. All participants provided informed consent prior to participation, and the study protocol (Protocol #XXX) was approved by the Carnegie Mellon University Institutional Review Board. All participants completed experiments using the hip exoskeleton, knee exoskeleton, and an IMU sensor suite.

B. Open-Source Training Data

We trained all biological joint moment estimators exclusively using publicly available biomechanics data (Fig. 1A). The aggregated database included level-ground, incline, and decline walking data from the datasets used in Camargo et al. [11], Scherpereel et al. [12], and Molinaro et al. [8], [9]. Acyclic tasks (i.e. squats) and stair-negotiation trials were excluded as they were out of the scope of the trials and .

Sagittal-plane joint angles and net joint moments were extracted from the processed files, and joint angular velocities were calculated from the filtered joint-angle trajectories using B-spline derivatives for 1-DoF joints (i.e. knee, ankle) and rotation matrix finite differences for 3-DoF joints (i.e. hip). Joint coordinate and moment conventions were harmonized to a common right-limb sagittal-plane convention (hip flexion positive, knee flexion negative, ankle dorsiflexion positive). Joint moments were normalized by participant body mass (Nm/kg). Source data recorded at approximately 200 Hz were decimated to a model sampling frequency of 100 Hz.

Participants were divided into mutually exclusive training (51), validation (2), and test (3) sets using a fixed random seed of 42. Input and output normalization were disabled for the evaluated configurations, meaning models received physical kinematic units and directly predicted mass-normalized joint moments.

C. Biological Joint Moment Estimator

We used a causal temporal convolutional network [19] to estimate biological joint moments from joint kinematics. The network consisted of five temporal convolutional blocks (80 hidden channels per block, kernel size of five, and exponentially increasing dilation factors of 1, 2, 4, 8, and 16). Each block included causal convolution, batch normalization, a rectified linear unit activation, and dropout with a probability of 0.1, resulting in approximately 292,000–294,000 trainable parameters depending on the input configuration. The network processed a sliding input window containing the preceding 100 samples (1.0 s at 100 Hz), ending at the current prediction time to ensure causality.

Models were trained via PyTorch [20], using the Huber loss (transition parameter of 0.5 Nm/kg) and the AdamW optimizer (weight decay 1×10^{-4} , batch size 512, gradient

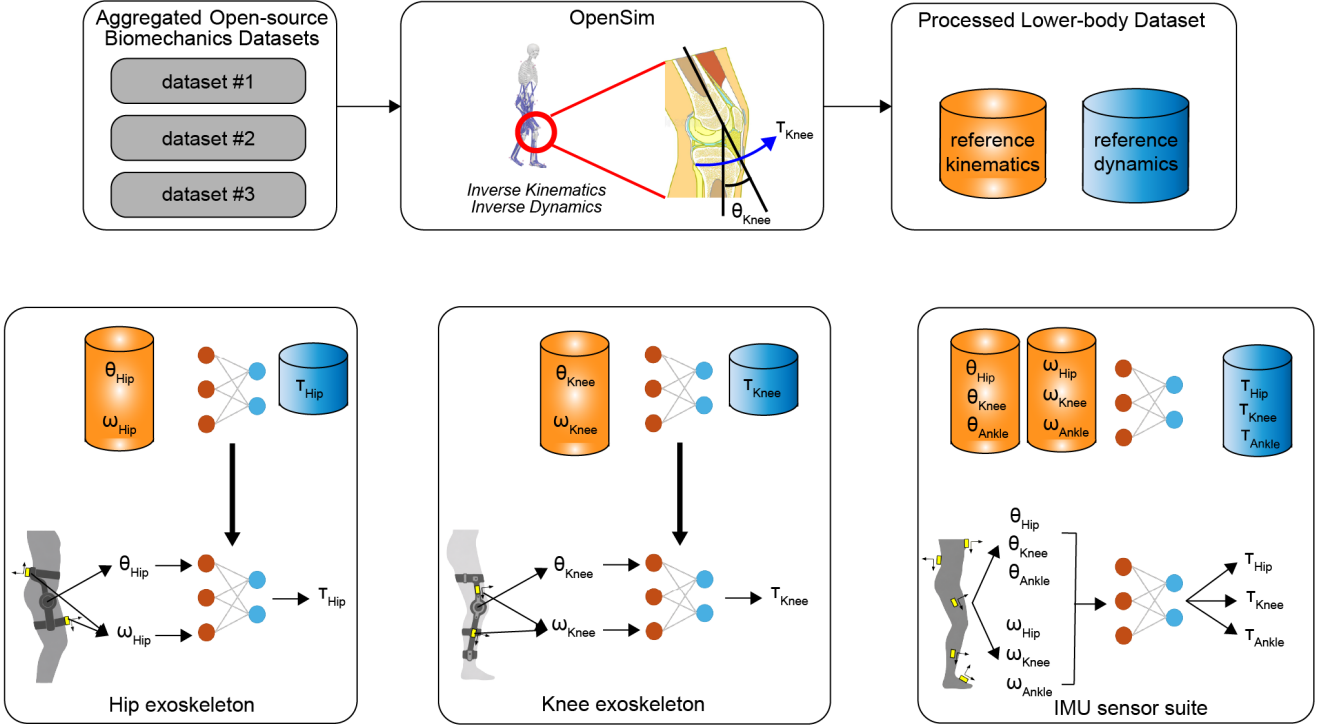


Fig. 1. Overview of the proposed kinematics-based framework for device-agnostic biological joint moment estimation. (A) The biological joint moment estimator is trained using reference kinematics and dynamics derived via OpenSim from aggregated open-source biomechanics datasets. The trained model is then deployed across diverse hardware configurations, including (B) a robotic hip exoskeleton utilizing hip joint kinematics (θ_{Hip} , ω_{Hip}) to predict hip moments (T_{Hip}), (C) a robotic knee exoskeleton utilizing knee joint kinematics (θ_{Knee} , ω_{Knee}) to predict knee moments (T_{Knee}), and (D) an IMU sensor suite utilizing multi-joint lower-limb kinematics to simultaneously predict hip, knee, and ankle moments.

clipping norm 1.0). A cosine-annealing scheduler reduced the primary learning rate from 2×10^{-5} to a minimum of 1×10^{-6} . Models were trained for up to five epochs, utilizing early stopping based on validation mean squared error with a patience of two epochs. Separate model configurations were trained for the single-joint inputs (hip or knee) and multi-joint inputs (bilateral hip, knee, and ankle). After training, the TCN blocks were exported in TensorRT format [21] to accelerate inference on the edge controller's GPU.

D. Wearable Sensing and Exoskeleton Platforms

1) *Robotic Hip Exoskeleton*: Hip-exoskeleton experiments utilized a bilateral device developed for sagittal-plane assistance [22], featuring AK80-9 brushless actuators with integrated motor encoders and an onboard Jetson-based computing system. Hip angle and angular velocity, derived from the encoders and relative movement of pelvis- and thigh-mounted IMUs, were provided to the shared single-joint hip moment estimator (Fig. 1B). Output torque for each leg was commanded as:

$$\text{clip} \left[\mathcal{F}_{\text{out}} \left(\alpha_{\text{hip}} m \hat{M}_{\text{bio}}(t - \Delta t_{\text{hip}}) \right), -\tau_{\text{lim}}, \tau_{\text{lim}} \right], \quad (1)$$

where $\hat{M}_{\text{bio}}$ is the predicted biological hip moment, m is participant body mass, $\alpha_{\text{hip}} = 0.12$ is the torque scaling factor, $\Delta t_{\text{hip}} = 120$ ms is the command delay, $\mathcal{F}_{\text{out}}$ represents the causal filtering operations, and $\tau_{\text{lim}} = 15$ N·m is the torque limit.

2) *Robotic Knee Exoskeleton*: Knee-exoskeleton experiments utilized a unilateral quasi-direct-drive device worn on the right leg. The actuated knee module (1.3 kg) and a waist-mounted electronics module (0.9 kg) communicated via a Jetson Orin and Teensy microcontroller at 100 Hz. Knee angle was obtained from the motor encoder, while angular velocity was estimated from thigh- and shank-mounted IMUs (Fig. 1C). The knee controller mirrored the structure of Eq. 1, but utilized a scaling factor of $\alpha_{\text{knee}} = 0.09$, no additional command delay, and a causal fourth-order 6-Hz low-pass filter on the commanded output.

3) *IMU Sensor Suite*: A wireless inertial sensing system [23] (MTw Awinda, Xsens, Netherlands) provided unassisted, bilateral, multi-joint kinematic information (Fig. 1D). Sensors were affixed to the sternum, posterior pelvis, and bilaterally on the thighs, shanks, and feet. Raw measurements were processed using a vector quaternion filter-based orientation and inverse-kinematics pipeline at 100 Hz. Joint angles were filtered using a fourth-order zero-phase Butterworth low-pass filter (6 Hz cutoff), and angular velocities (derived via centered finite-difference) were filtered at 15 Hz.

E. Experimental Protocol

Experiments were conducted on a force-instrumented split-belt treadmill (Bertec, USA). Participants performed level-ground walking (LG) at 0.8, 1.2, and 1.6 m/s, ramp ascent (RA, $+10^\circ$) at 0.8 m/s, and ramp descent (RD, -10°) at

figures/hardware.png

Fig. 2. Overview of the hardware.

0.8 m/s. Each trial lasted approximately 80 s, with the first and last 10 s trimmed to isolate steady-state gait. We analyzed 31 hip exoskeleton trials, 16 knee exoskeleton trials, and 39 IMU sensor suite trials, excluding specific 0.8 m/s and 1.6 m/s LG trials due to corrupted reference OpenSim data.

F. Reference Biomechanics and Moment Estimation

Reference biomechanics were collected at 100 Hz using a 24-marker optical motion-capture system (Vicon, UK). Participant-specific OpenSim 4.5 [24] musculoskeletal models (based on standard lower-extremity models [25]–[27]) were

scaled for each session, directly incorporating the added mass and inertia of the hip (2.0 kg pelvis, 0.5 kg per femur, 1.3 kg torso) or knee (0.9 kg pelvis, 0.91 kg right femur, 0.39 kg right tibia) exoskeletons. Inverse kinematics coordinates were low-pass filtered at 6 Hz before calculating net sagittal-plane joint moments via inverse dynamics. For assisted trials, biological joint moments were estimated by aligning and subtracting the measured exoskeleton motor torque from the OpenSim-derived net joint moment prior to mass normalization.

G. Synchronization and Model Evaluation

Exoskeleton and motion-capture systems were synchronized via TTL pulses generated at the start and end of each trial. OpenSim reference signals were linearly interpolated onto the 100-Hz exoskeleton time base. IMU sensor suite data were synchronized offline using cross-correlation of filtered bilateral hip and knee angles.

Model performance was evaluated using mass-normalized Root Mean Square Error (RMSE) and the coefficient of determination (R^2) between 6-Hz low-pass filtered predicted and reference moments. Summary values were calculated by assigning equal weight to each retained trial-channel combination. For the supplementary IMU analysis, multi-joint and per-joint models were statistically compared using paired one-sided Wilcoxon signed-rank tests with Holm–Bonferroni correction ($\alpha = 0.05$).

III. RESULTS

Unless otherwise specified, results are reported as mean $\pm$ standard deviation. The hip-exoskeleton analysis included 31 trials; the AB05 LG 1.6-m/s trial was excluded because of an incorrect trial identifier. The knee-exoskeleton analysis included all 16 collected trials. The IMU sensor suite analysis included 39 trials; the AB05 LG 0.8-m/s trial was excluded because of an incorrect trial identifier. Detailed joint moment prediction performance metrics across all joints, conditions, and platforms are summarized in Table I.

A. Biological Joint Moment Estimation Across Wearable Platforms

The kinematics-based estimators successfully predicted biological joint moments across all three wearable platforms (Table I). Utilizing hip-exoskeleton kinematics, knee-exoskeleton kinematics, and multi-joint IMU-derived kinematics, the estimators achieved robust predictive performance across single-joint localized sensing and multi-joint whole-limb inertial kinematics, without requiring target-device training or fine-tuning.

B. Hip Exoskeleton Moment Estimation

Hip moment estimation performance utilizing localized, onboard sensing varied depending on the walking condition. As detailed in Table I, absolute RMSE increased with level-ground walking speed, reaching its peak during the 1.6 m/s condition. However, the condition-averaged R^2 remained consistently strong across all evaluated speeds and during ramp ascent, indicating that the estimator successfully captured the temporal structure of the hip moment profile even as absolute moment magnitudes increased.

C. Knee Exoskeleton Moment Estimation

Using knee angle obtained from the motor encoder and knee angular velocity derived from the thigh and shank IMUs, the estimator successfully predicted knee moments during both ramp conditions. Performance was generally lower during ramp descent than during ramp ascent (Table I).

To evaluate the influence of the device-derived kinematic inputs, the same model was additionally evaluated using knee angles and angular velocities obtained directly from Vicon inverse kinematics. Relative to the exoskeleton-derived inputs, utilizing Vicon-derived inputs reduced the mean RMSE by 20.2% and increased the mean R^2 by 0.125 across both ramp conditions. Notably, performance remained lower during ramp descent than during ramp ascent even when using Vicon-derived kinematics.

D. Multi-Joint Moment Estimation Using IMU Sensor Suite Kinematics

The IMU-based estimator predicted bilateral hip, knee, and ankle moments across all evaluated locomotor conditions using whole-limb inertial kinematics. Condition-averaged R^2 increased with level-ground walking speed and remained strong during both ramp ascent and ramp descent (Table I). Across all predictions, the ankle moment estimates exhibited the highest mean R^2 on both sides, whereas the left-knee predictions exhibited the lowest mean R^2 and the greatest between-trial variability.

E. Comparison of Multi-Joint and Single-Joint IMU Sensor Suite Results

The multi-joint IMU-derived model was compared with separate single-joint models across 234 paired joint-trial observations. The multi-joint model demonstrated superior performance, reducing the overall mean RMSE by 17.2% compared to the single-joint models. The single-joint models produced higher RMSE in 73.5% of the paired observations and lower R^2 in 73.9% of the observations.

Paired one-sided Wilcoxon signed-rank tests confirmed these improvements, indicating significantly lower RMSE for the multi-joint model at the hip ($p = 0.007$), knee ($p < 0.001$), and ankle ($p < 0.001$). The multi-joint model also produced significantly higher R^2 values at all three joints (all $p < 0.001$), demonstrating the clear benefit of adjacent-joint kinematic context.

F. Accuracy of IMU-Derived Joint Kinematics

Across the bilateral hip, knee, and ankle channels, the IMU-derived joint angles demonstrated high agreement with the corresponding Vicon inverse-kinematics trajectories (R^2 of 0.919 ± 0.103), with an overall RMSE of $7.438 \pm 4.320^\circ$. Hip and knee angles exhibited the strongest correlation ($R^2 > 0.96$), whereas ankle-angle agreement was comparatively lower ($R^2 \approx 0.80$), contributing to the observed variability in downstream moment prediction.

IV. DISCUSSION

This study evaluated joint kinematics as a common intermediate representation for biological joint moment estimation across three distinct wearable sensing platforms. Models trained exclusively on open-source biomechanics data were applied—without target-device training or fine-tuning—to kinematics obtained from a bilateral hip exoskeleton, a unilateral

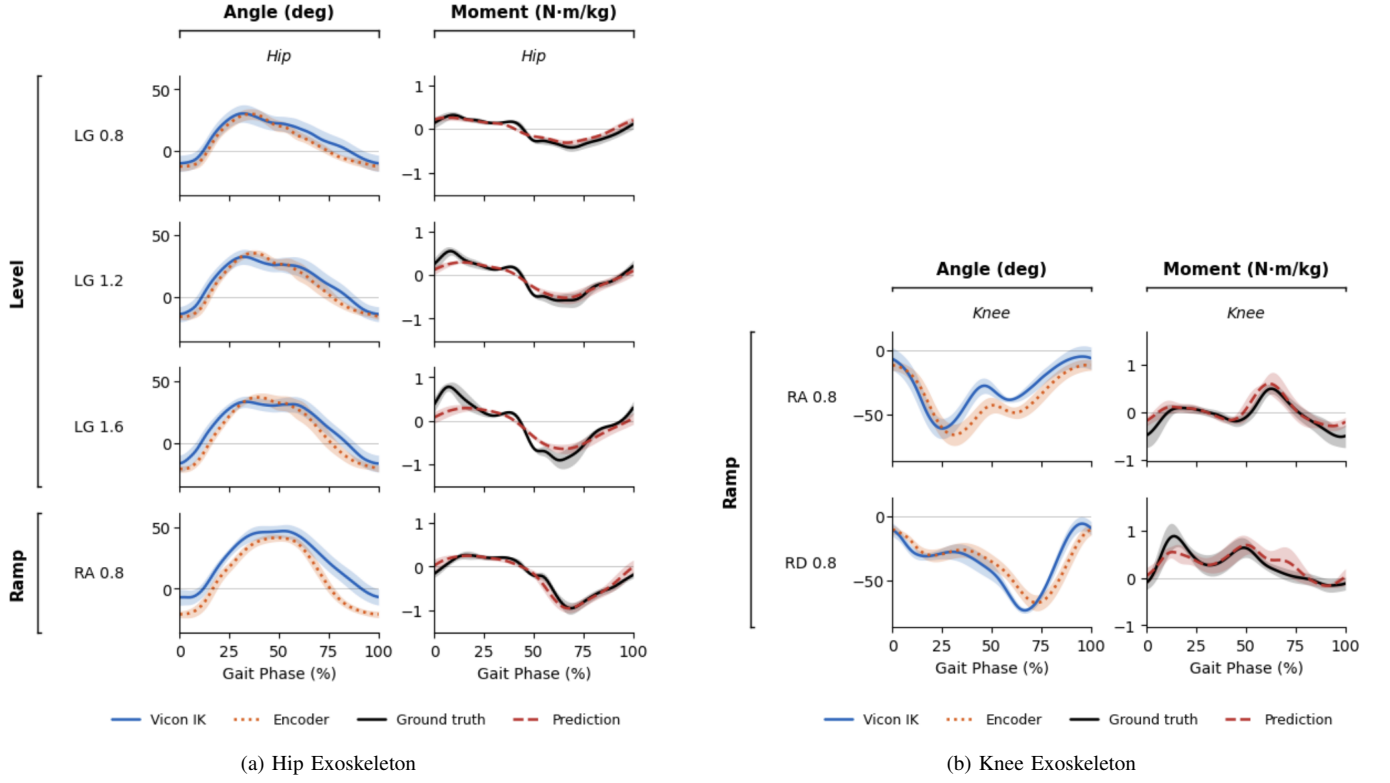


Fig. 3. Time-series kinematics and predicted biological joint moments for (a) the hip exoskeleton across level-ground and ramp conditions and (b) the knee exoskeleton during ramp ascent and descent. The shaded regions denote one standard deviation.

TABLE I
JOINT MOMENT PREDICTION PERFORMANCE ACROSS JOINTS AND TRIAL CONDITIONS.

Trial	Joint	Condition	RMSE (Nm/kg)	R^2
Hip exo	Hip	LG 0.8	0.12 ± 0.04	0.76 ± 0.13
	Hip	LG 1.2	0.16 ± 0.03	0.81 ± 0.08
	Hip	LG 1.6	0.25 ± 0.08	0.76 ± 0.10
	Hip	RA 0.8	0.16 ± 0.03	0.84 ± 0.09
Knee exo	Knee	RA 0.8	0.17 ± 0.03	0.65 ± 0.14
	Knee	RD 0.8	0.22 ± 0.03	0.58 ± 0.14
IMU sensor suite	Hip	LG 0.8	0.11 ± 0.02	0.80 ± 0.07
	Hip	LG 1.2	0.14 ± 0.02	0.88 ± 0.04
	Hip	LG 1.6	0.16 ± 0.02	0.92 ± 0.02
	Hip	RA 0.8	0.21 ± 0.06	0.83 ± 0.06
	Hip	RD 0.8	0.13 ± 0.03	0.75 ± 0.10
	Knee	LG 0.8	0.13 ± 0.07	0.60 ± 0.25
	Knee	LG 1.2	0.11 ± 0.04	0.83 ± 0.08
	Knee	LG 1.6	0.12 ± 0.04	0.90 ± 0.05
	Knee	RA 0.8	0.14 ± 0.05	0.78 ± 0.09
	Knee	RD 0.8	0.17 ± 0.10	0.82 ± 0.20
	Ankle	LG 0.8	0.16 ± 0.07	0.92 ± 0.05
	Ankle	LG 1.2	0.14 ± 0.05	0.96 ± 0.02
	Ankle	LG 1.6	0.15 ± 0.05	0.97 ± 0.03
	Ankle	RA 0.8	0.19 ± 0.04	0.90 ± 0.04
	Ankle	RD 0.8	0.16 ± 0.03	0.91 ± 0.03

Values are reported as mean $\pm$ standard deviation across participants. LG = level ground; RA = ramp ascent; RD = ramp descent. Walking speeds are reported in m/s. RMSE was normalized by participant body mass and is reported in Nm/kg. Exoskeleton results were calculated using the deployed encoder and IMU-derived kinematic inputs. For the IMU sensor suite trials, left- and right-side metrics were averaged within each participant before calculating group-level means and standard deviations. Rows in bold indicate best-performing condition based on mean R^2 .

Table I, the estimators successfully predicted joint moments across all three configurations. These findings demonstrate that joint kinematics can effectively decouple hardware-specific sensing from downstream biological joint moment estimation.

The results do not seek to establish the superiority of one sensing platform over another; rather, they highlight the versatility of a shared, kinematics-based formulation across disparate hardware configurations and deployment environments. The exoskeleton models operated using local signals from onboard encoders and IMUs, whereas the IMU sensor suite model utilized bilateral multi-joint kinematics processed offline. Although differences in input dimensionality, filtering, and intended use preclude a direct comparison of sensing-platform accuracy, successful estimation across all three configurations indicates that wearable systems do not need to share the same raw sensor modalities or placements to leverage estimators trained from a common biomechanical data source.

A. Joint Kinematics as a Common Representation Across Wearable Platforms

Physiological-state estimators for exoskeleton control are conventionally trained using direct signals from device-mounted encoders, IMUs, pressure sensors, or other wearables. While these signals may provide accurate predictions for the specific device on which the data were collected, their distributions are highly dependent on sensing hardware, sensor placement, coordinate definitions, calibration, and the preprocessing pipeline. Consequently, models trained directly from these raw signals are difficult to transfer to devices with

knee exoskeleton, and an IMU sensor suite. As detailed in

different sensing configurations.

The proposed framework mitigates this dependency by decoupling the estimation pipeline into two distinct stages. First, each wearable device uses its available sensors to measure or estimate joint kinematics. Second, a downstream model maps these kinematics to biological joint moments. Under this formulation, the hardware-specific component is responsible solely for producing joint angles and angular velocities in a common coordinate system, allowing the downstream estimator to operate in a shared, hardware-agnostic representation.

This separation was validated using substantially different sensing configurations. The hip exoskeleton derived hip angle from its motor encoder and hip angular velocity from the relative movement of pelvis- and thigh-mounted IMUs. Similarly, the knee exoskeleton provided knee angle from its motor encoder and knee angular velocity from thigh- and shank-mounted IMUs. In contrast, the IMU sensor suite supplied bilateral hip, knee, and ankle kinematics via an inertial motion-capture pipeline. Despite these structural differences, all three systems provided compatible inputs that were successfully processed by models trained on open-source joint kinematics and inverse-dynamics moments.

The exoskeleton results are particularly relevant to real-time control, as the estimator relied exclusively on local, onboard sensing. Condition-averaged R^2 values remained strong across level-ground and ramp-ascent walking. While absolute RMSE increased with level-ground walking speed, the corresponding R^2 remained consistent. This indicates that the estimator successfully captured the temporal structure of the hip moment profile at higher speeds, with the increased absolute error driven primarily by the larger magnitude of the reference moments.

The IMU sensor suite results extended this principle to bilateral, multi-joint biomechanics estimation. The estimator maintained high condition-averaged R^2 values during ramp walking and peaked during fast level-ground walking. These findings confirm that a kinematics-domain estimator is equally effective when utilizing a richer, whole-limb representation, such as data gathered during offline wearable biomechanics data collection.

B. Influence of Kinematic Input Quality

Although joint kinematics provide a more transferable input representation than raw sensor signals, prediction performance remains inherently tied to the quality of the kinematic estimates. This dependence was most evident in the knee-exoskeleton analysis. When the estimator was evaluated using Vicon inverse-kinematics inputs rather than encoder- and IMU-derived kinematics, RMSE decreased by approximately 20% and R^2 improved notably.

Because the downstream model remained identical, the performance gains achieved with Vicon-derived inputs suggest that a portion of the knee-exoskeleton prediction error stemmed from inaccuracies in the wearable-derived kinematics. Encoder measurements can deviate from true biological joint motion due to actuator-joint misalignment, interface compliance, and relative movement between the exoskeleton

and the user [28], [29]. Similarly, deriving joint angular velocity from segment-mounted IMUs can introduce errors related to sensor placement, soft-tissue artifact, and attachment variability.

Notably, knee moment estimation was less accurate during ramp descent than during ramp ascent across both input sources. The persistence of this performance gap with Vicon inputs suggests that the difficulty of modeling ramp descent was not solely an artifact of the exoskeleton sensors. Ramp descent may involve distinct joint moment patterns or participant-specific movement strategies that are under-represented in the open-source training data, though further biomechanical analysis is required to isolate these factors.

The IMU sensor suite kinematic comparison further demonstrated that high waveform agreement does not guarantee low absolute joint-angle error. Across all joints, IMU-derived angles agreed strongly with Vicon inverse kinematics in terms of R^2 , yet exhibited notable absolute RMSE. Hip and knee angles exhibited higher agreement, whereas ankle-angle agreement was comparatively lower. This suggests that the estimator is somewhat robust to offset or amplitude discrepancies provided the overall temporal structure of the kinematics is preserved. Nonetheless, refining wearable kinematic estimation remains a clear pathway to improving downstream biological joint moment prediction.

C. Benefits of Multi-Joint Kinematic Context

Comparing the multi-joint and single-joint IMU sensor suite results revealed that biological joint moment estimation benefits significantly from the kinematic context of adjacent joints. Across all paired observations, the multi-joint model reduced mean RMSE by over 17% and consistently increased mean R^2 . Improvements were consistent across the hip, knee, and ankle, with paired tests indicating significantly lower RMSE and higher R^2 for the multi-joint model at each joint.

These improvements reflect the highly coordinated nature of lower-limb movement. The moment generated at any given joint depends not only on its local kinematics but also on whole-limb configuration, intersegmental dynamics, ground contact state, and the phase of the gait cycle. Multi-joint kinematics provide implicit information about these broader mechanical states that local joint motion alone cannot capture.

The contrast between the IMU sensor suite and exoskeleton results highlights a practical design tradeoff: multi-joint inputs enhance prediction accuracy but require a more extensive sensor network. Conversely, local single-joint models are less information-rich but can operate seamlessly using an exoskeleton's existing sensors. The optimal formulation therefore depends on the specific application. A standalone biomechanics tracking system may readily support multi-joint estimation, whereas an exoskeleton with strict hardware limitations may prioritize local, causal estimation.

Crucially, this finding outlines a scalable pathway for improving exoskeleton estimators without requiring full-body motion capture. Wearable context can be enriched incrementally by incorporating sensors from neighboring joints, foot-contact states, or other locally available signals. The

kinematics-based framework does not require identical input dimensionality across all platforms; rather, it gracefully accommodates varying levels of biomechanical context.

D. Implications for Transferable Exoskeleton Control

The proposed framework addresses two primary barriers to deploying biological joint moment estimators across diverse wearable devices. First, an estimator developed on one device’s raw signals often requires inputs that are unavailable or fundamentally different on a new device. Second, open-source biomechanics datasets rarely include the raw sensor signals corresponding to every possible wearable configuration. Consequently, models trained on hardware-specific inputs typically require exhaustive de novo data collection for each new target device.

Using joint kinematics as an intermediate representation bypasses both bottlenecks. A new device does not need to emulate the original training system’s raw sensors; it only needs to generate the joint angles and angular velocities expected by the downstream model. This allows developers to leverage massive open-source datasets (containing kinematics and inverse-dynamics moments) for model training, even if those datasets completely lack the target device’s specific encoders or IMUs.

While this approach drastically reduces data collection burdens, it does not entirely eliminate device-specific development. Each platform still requires careful calibration, coordinate harmonization, signal processing, and validation of its kinematic estimates. However, this is a much narrower technical requirement than orchestrating a full data collection protocol with synchronized wearable signals, motion capture, ground-reaction forces, and reference biological joint moments. The mapping from joint kinematics to biological joint moments becomes a reusable module, effectively isolating the hardware-specific engineering to the task of producing compatible kinematics.

For real-time exoskeleton control, the hip and knee experiments confirm that this estimator can be embedded with local device sensing for causal inference. For wearable biomechanics data collection, the IMU sensor suite results validate that the same core principles support bilateral, multi-joint offline estimation. Together, these use cases demonstrate that kinematics-based models hold significant promise as modular, reusable estimation components across the wearable robotics and biomechanics ecosystem.

E. Study Impact and Limitations

The primary contribution of this study is demonstrating that biological joint moment estimators—trained entirely on open-source biomechanics data—can generalize to multiple wearable sensing platforms without requiring target-device data for training or fine-tuning. This framework stands to significantly reduce the data-collection burden when developing physiological-state estimators for new exoskeletons while simultaneously amplifying the value of existing open-source biomechanics datasets. Furthermore, the results highlight two actionable pathways for performance improvement: refining

the accuracy of device-derived joint kinematics, and integrating broader biomechanical context when hardware permits.

Several limitations must be considered. First, both evaluated exoskeletons utilized quasi-direct-drive actuation and were developed within the same laboratory. While they differed in joint targeted, sensing configuration, and laterality (unilateral vs. bilateral), they inevitably share underlying design philosophies, control architectures, interface strategies, and signal-processing conventions. These results do not yet guarantee transferability to exoskeletons with radically different actuation paradigms or transmission architectures developed by other institutions. Future validation should encompass commercially available systems and research hardware from independent laboratories.

Second, the exoskeleton experiments were conducted at relatively conservative assistance magnitudes, with applied torques generally peaking between 5–7 N·m. This conservative level was intentionally selected to evaluate the estimator during active assistance while minimizing sensor noise and interface shifts induced by heavy actuator operation. While the estimators maintained robust performance under these conditions, the devices are capable of generating up to approximately 18 N·m. Higher torque levels may induce more pronounced changes in user gait, increased interface deformation, greater human–robot misalignment, and amplified vibrational artifacts in onboard sensors. Future work must verify estimation performance across the full operational bandwidth of these devices.

Finally, the models were trained and evaluated exclusively on periodic walking tasks (level-ground, ramp-ascent, and ramp-descent). The current architecture does not encompass stair ambulation, gait transitions, obstacle negotiation, or turning. Expanding the training database to include these acyclic and complex behaviors is necessary to broaden the model’s locomotor repertoire. The architecture could be further enhanced by incorporating task-specific output heads or contextual variables that encode the user’s current locomotor state. Non-inertial sensors, such as pressure insoles, could provide critical insight into contact timing, limb loading, and terrain interactions that joint kinematics alone cannot fully capture. These extensions remain critical frontiers for future research and should not be assumed to generalize directly from the steady-state walking conditions evaluated here.

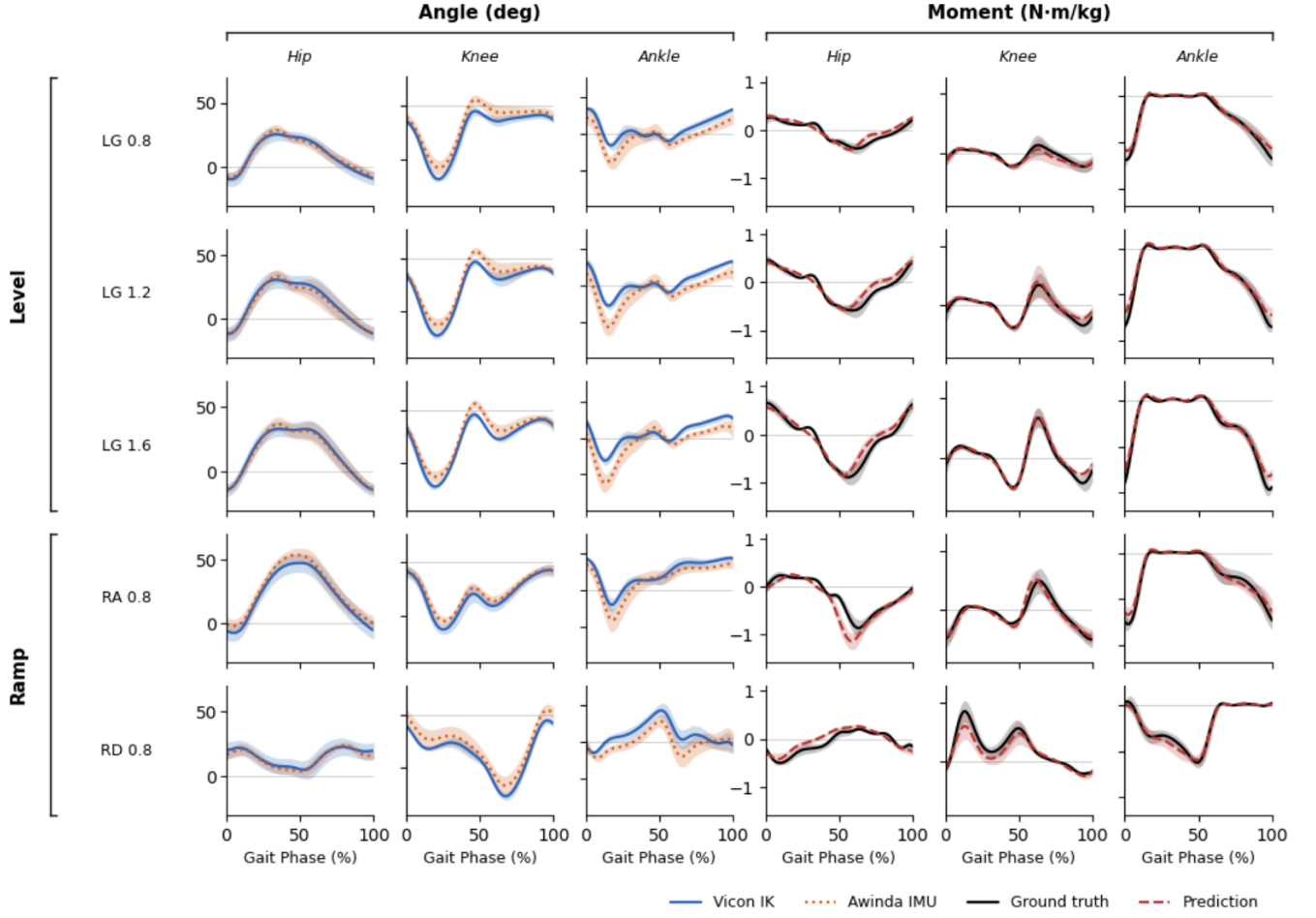
V. CONCLUSION

This study demonstrated that joint kinematics can serve as an effective, hardware-agnostic intermediate representation for estimating biological joint moments across diverse wearable sensing platforms. By training models exclusively on open-source biomechanics datasets, kinematics-based estimators successfully predicted joint moments using both localized, exoskeleton-derived kinematics and whole-limb, IMU-derived kinematics—without requiring any target-device training or fine-tuning. These findings validate a modular approach to physiological-state estimation, effectively decoupling downstream moment prediction from the specific sensor configurations of the deployment hardware. Ultimately, this framework provides a scalable pathway to reduce the data-collection

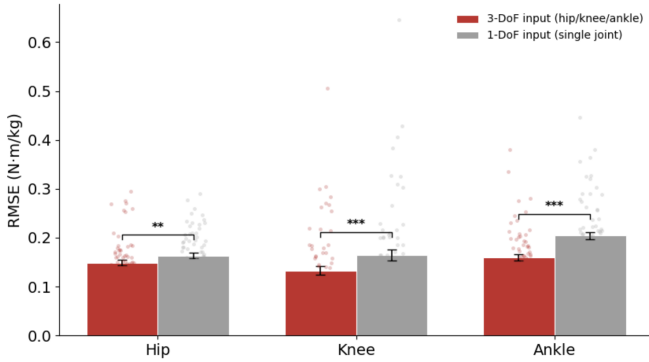
burden when developing controllers for new wearable devices. Future work will focus on integrating a broader range of exoskeleton datasets, evaluating the benefits of limited-data fine-tuning to improve performance on specific devices, and further validating this kinematics-based framework during real-time, closed-loop assistance of other devices.

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(a) Awinda Time-Series Kinematics and Moments



(b) Multi-Joint vs. Single-Joint RMSE

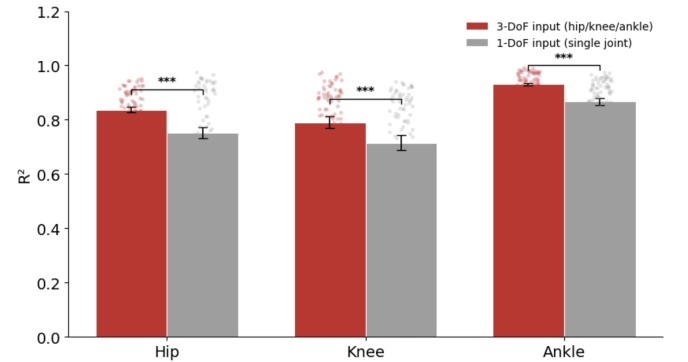
(c) Multi-Joint vs. Single-Joint R^2

Fig. 4. Evaluation of the kinematics-based framework using the Awinda IMU sensor suite. (a) Time-series kinematics and predicted moments across level-ground and ramp walking conditions. Comparisons of (b) RMSE and (c) R^2 demonstrate the significant benefit of utilizing multi-joint kinematic context (3-DoF) over isolated single-joint inputs (1-DoF) (** $p < 0.01$, *** $p < 0.001$).